incentivize registrants to take additional care to ensure the accuracy of the disclosures, thereby resulting in more reliable disclosures.[2751] Several commenters expressed support for treating climate-related disclosures as filed, noting that it would help improve investor confidence in the accuracy and completeness of such disclosures.[2752]

For disclosures other than financial statement disclosures, the final rules will provide registrants with the flexibility to determine the appropriate placement within their filing of climate-related disclosures. While this could affect investors' ability to easily locate and compare those disclosures, we believe that this concern is largely mitigated by the final rules' structured data requirement. The structured disclosure requirements we are adopting, including the requirement to tag such disclosures using XBRL, will enable search and retrieval of the disclosures on an automated and large-scale basis, allowing investors, and the market, to process information much more effectively and efficiently as compared to manual searches through unstructured formats. This will improve investors' assessment of companies' estimated future cash flows, leading to more accurate company valuations and lowering companies' cost of capital.[2753]

Additionally, having access to more reliable information could result in cost savings for those investors who collect or organize information about climate-related risks. Several

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[2751] However, we note that these benefits will be mitigated for certain forward-looking statements, including those related to transition plan disclosures, scenario analysis, internal carbon pricing, and climate-related targets and goals, as these statements will have the benefit of safe harbor protections if the safe harbor requirements are satisfied.

[2752] *See, e.g.*, letters from Amer. For Fin. Reform, Sunrise Project *et al*.; Ags of Cal. *et al*.; CalPERS; Ceres; CFA; Engine No. 1; Franklin Templeton; PwC; SKY Harbor; and TotalEnergies.

[2753] *See* Christensen *et al*. (2021), at 1187; R. Lambert, C. Leuz & R.E. Verrecchia, *Accounting Information, Disclosure, and the Cost of Capital*, 45 J. of Acct. Rsch. 385 (May 2007); D. Easley & M. O'Hara, *Information and the Cost of Capital*, 59 J. of Fin 1553 (Aug. 2004); R. Lambert, C. Leuz & R.E. Verrecchia, *Information Asymmetry, Information Precision, and the Cost of Capital*, 16 Rev. of Fin. 1 (Jan. 2012).

commenters emphasized the scale of the resources required to render the currently available information on climate-related disclosures useful to their decisions.[2754]

Similarly, investors also may benefit from the final rules if the required disclosures change the nature and degree to which investors rely on third parties that provide ESG ratings or scores. To the extent there is overlap between the disclosures required by the final rules and the types of information considered by ESG ratings providers, the final rules may reduce reliance on these third parties, thereby reducing costs incurred by investors to obtain decision-useful information. ESG ratings are not necessarily standardized or transparent with respect to their underlying methodologies, and several studies have found that different ESG ratings providers often assign inconsistent ratings for the same registrant.[2755] To the extent the final rules reduce reliance on these ratings, registrants and investors could benefit by saving money that would otherwise be spent on obtaining third-party ESG ratings.[2756] Alternatively, the disclosures elicited by the final rules may increase the value of these third-party services to the extent that the third-party services are able to leverage the enhanced disclosures to provide investors with

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[2754] *See, e.g.*, letter from CalSTRS (stating, "The current reporting requirements are insufficient for investors to assess corporate climate risk and the related financial impacts to execute investment decisions. CalSTRS spends approximately $2,200,000 per year to access climate research, analyze available data, and develop methods to estimate climate risks and opportunities for assets in our portfolio. In addition to two full-time investment staff members, CalSTRS consults external advisors to learn how other global asset owners determine climate risk exposures to their portfolios given the lack of reliable, consistent, and comprehensive data. A conservative estimate of the variable cost of these combined human resources is $550,000 annually.").

[2755] Florian Berg, Julian Kölbel & Roberto Rigobon, *Aggregate Confusion: The Divergence of ESG Ratings*, 26 Rev. Fin. 1315 (Nov. 2022). The authors found that the correlations between six different ESG ratings are on average 0.54, and range from 0.38 to 0.71, while the correlations between credit ratings were 0.99. *See also* Scott Robinson *et al*., *supra* note 2721; Dane Christensen, George Serafeim & Anywhere Sikochi, *Why is Corporate Virtue in the Eye of the Beholder? The Case of ESG Ratings*, Acct. Rev. (Feb. 26, 2021), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3793804 (retrieved from SSRN Elsevier database).

[2756] Because ESG ratings encompass information beyond climate-related matters, registrants and investors may still obtain ESG ratings for reasons unrelated to climate-related information.

greater market insights. The disclosures may also allow registrants to better monitor ESG ratings, which could reduce the risk of greenwashing.

b. Costs

The final rules will impose direct costs of compliance on registrants. We use the term "direct costs" or "compliance costs" to include (1) any costs related to developing or maintaining systems for collecting information to comply with the final rules, (2) costs of preparing and presenting the resulting disclosures for Commission filings, which we refer to as "reporting costs,"[2757] (3) costs associated with assuring the accuracy of the disclosures, such as audit and attestation costs, and (4) any legal or disclosure review costs incurred to support management's assertion that the disclosures comply with the final rules. These costs could be incurred internally (e.g., through employee hours or hiring additional staff) or externally (e.g., via third-party service providers, such as auditors or consultants). Numerous commenters expressed concerns over the direct costs of compliance on registrants of the proposed rules.[2758] As discussed in section II.A, the final rules include certain modifications relative to the proposed rules that reduce overall costs and help address commenters' concerns about the time and resources required to comply with the final rules' requirements.[2759] This concern could further be mitigated for certain registrants to the extent that the final rules generally align with the disclosure frameworks that they are already using for their voluntary disclosures or disclosures

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[2757] Specifically, "reporting costs" refer to the costs of preparing information to be presented in Commission filings, separate from any prior costs or resources expended in obtaining or developing such information. For example, the final rules will require some registrants to disclose their Scope 1 and 2 emissions. For registrants that already disclose them, the incremental cost will only be the reporting cost, distinct from any costs they have previously voluntarily incurred related to developing emissions measurement/estimation systems and processes in order to quantify their emissions.

[2758] *See, e.g.*, letters from Chamber; Nutrien; Williams Companies; Energy Transfer LP; Hess; PPL; NRF; RILA; ConocoPhillips; NASDAQ; API; and SCG.

[2759] *See, e.g.*, letter from Chamber.

that are, or will be, required by state, Federal, or other laws.[2760] Many commenters submitted cost estimates for the proposed rules that varied considerably depending on a given company's size, industry, complexity of operations, and other characteristics. We review these comments and discuss cost estimates in detail in sections IV.C.2 and IV.C.3. The remainder of this section focuses on other costs that may result from the final rules.

The final rules may result in additional litigation risk for registrants.[2761] However, the final rules include several changes from the proposal to mitigate these concerns. For example, certain forward-looking statements, including those related to transition plan disclosures, scenario analysis, internal carbon pricing, and climate-related targets and goals, will have the benefit of certain liability protections if the adopted safe harbor requirements are satisfied. Another example is the inclusion of phase in periods after the effective date to provide registrants with additional time to become familiar with and meet the final rules' disclosure requirements. In addition, Scope 3 emissions disclosure is no longer required and the amendments to Regulation S-X have been modified to lessen the compliance requirements, among other examples.

2760 *See* section IV.A.3 for discussion of existing state and Federal laws.

2761 *See* letter from NAM (expressing concern about treating climate-related disclosures as "filed," noting the "evolving and uncertain nature" of GHG emissions disclosures could make it difficult for registrants to reach the degree of certainty necessary to assume the liability burden associated with reports filed with the Commission); *see also* letter from Chamber (noting that the complexity of the proposed rules could increase the likelihood of nuisance lawsuits that are intended to extract a settlement thereby increasing the cost of compliance with the rules). This commenter also pointed out that audit costs could increase if auditors were also subject to increased litigation risk. *See also* letter from Cunningham *et al*. (noting that "The SEC recognizes that a major cost of the Proposal concerns litigation risk."); Overdahl exhibit to letter from Chamber (noting the increase in litigation risk can also result in higher insurance costs for registrants and auditors).

Some commenters expressed concerns that the proposed disclosures would over-emphasize climate risks relative to other types of risks that investors might find important.[2762] A related concern that commenters raised is that potentially voluminous disclosures could obscure the information that investors deem most relevant to their investment or voting decisions.[2763] To mitigate such concerns (in addition to concerns related to the compliance costs) the final rules are less prescriptive in certain places relative to the proposed rules. The final rules also have additional materiality qualifiers such that registrants that determine climate risks to be immaterial will have fewer disclosure obligations relative to the proposal. These costs also are expected to be mitigated by the structured data requirements of the final rules, which will make it easier for investors to find and analyze relevant information in filings. These changes also address other commenters' concerns that mandatory climate disclosure requirements that are too prescriptive or granular may lead to inefficient changes in business strategies and limit or halt innovation in the market for voluntary climate disclosures.[2764]

[2762] *See, e.g.*, Overdahl exhibit to letter from Chamber, stating, "because climate-related information is just one factor among many other (potentially more relevant) factors, climate-related information is often not material;" *see also* letters from API; Western Energy Alliance and the U.S. Oil & Gas Association; Matthew Winden; American Council of Engineering Companies; Chamber; and Wisconsin Manufacturers & Commerce.

[2763] *See, e.g.*, letters from Chamber; Soc. Corp. Gov.; and ConocoPhillips. For example, investors may be unable to review all potentially relevant information, resulting in suboptimal decisions. *See, e.g.*, H.A. Simon, *A Behavioral Model of Rational Choice*, 69 Q. J. of Econ. 99 (Feb. 1955); H.A. Simon, *Rationality As Process and As Product of Thought*, 68 Am. Econ. Rev. 1 (May 1978); K.L. Chapman, N. Reiter & H.D. White, *et al*., *Information Overload and Disclosure Smoothing*, 24 Rev Acct. Stud. 1486 (Dec. 2019).

[2764] *See*, *e.g.*, letters from API; Matthew Winden; Footwear Distributors & Retailers of America; Petrol. OK.; and Chamber.

We also acknowledge the concerns expressed by several commenters that the proposed rules would have required the disclosure of confidential or proprietary information,[2765] which can put affected registrants at a competitive disadvantage. This consequence could alter registrants' incentives to develop strategies to manage climate-related risks where it would otherwise be beneficial to do so.[2766] The final rules have been narrowed relative to the proposed rules to provide additional flexibility to limit costs associated with the disclosure of competitively sensitive information, while retaining disclosures that will help investors understand registrants' climate-related risks. In particular, we have eliminated certain prescriptive requirements from the proposal that commenters identified as potentially revealing competitively sensitive information and for which the benefits to investors were less apparent.[2767] For example, by providing registrants with flexibility to determine how best to describe their strategy towards managing climate-related risks, the final rules may enable them to avoid disclosure of competitively sensitive information.[2768] Furthermore, while we have eliminated the requirement to disclose ZIP code level information, the final rules continue to require location disclosures sufficient to understand a registrants' exposure to physical risks. We also have eliminated the requirement to disclose interim targets or goals. At the same time, we acknowledge that, in some instances, a more flexible approach may also result in less

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[2765] *See, e.g., supra* notes 479, 596, and surrounding text. Proprietary costs are generally relevant for reporting that involves information about a companies' business operations or production processes and disclosures that are specific, detailed and process-oriented. *See, e.g.,* C. Leuz, A. Triantis & T.Y. Wang, *Why Do Firms Go Dark? Causes and Economic Consequences of Voluntary SEC Deregistrations*, 45 J. of Acct. & Econ. 181 (Aug. 2008); D.A. Bens, P.G. Berger & S.J. Monahan, *Discretionary Disclosure in Financial Reporting: An Examination Comparing Internal Firm Data to Externally Reported Segment Data*, 86 Acct. Rev. 417 (March 2011).

[2766] *See* section IV.D.

[2767] *See*, *e.g.*, letter from Business Roundtable (June 17, 2022).

[2768] *See*, *e.g.*, discussion in section II.C.1.c

comparable disclosures. While this has the possibility of reducing the value of the disclosures to investors, we believe this approach appropriately balances investor protection with concerns raised by commenters.

Relatedly, the final rules may have indirect cost implications for third-party service providers, such as ESG ratings providers. For example, the increased disclosures may reduce institutional investors' reliance on ESG ratings providers, which could negatively impact these providers.[2769] Conversely, more comprehensive disclosures could reduce the cost of producing ESG ratings or may improve the informational content of the ratings, thereby increasing demand. This could benefit not only the ratings providers, but also investors that rely on ESG ratings.

Many commenters raised concerns about costs to third parties from the proposed rules,[2770] with one commenter stating that "measuring and reporting of GHG emissions would be a prerequisite for doing business with registrants and most retailers under this proposal."[2771] Compared to the proposed rules, the final rules do not impose such costs because they do not include Scope 3 disclosure requirements. Other disclosure items under the final rules may continue to result in registrants seeking input from third parties, such as those disclosure items requiring disclosure of material impacts from climate-related risks on purchasers, suppliers, or other counterparties to material contracts with registrants. However, the final rules limit the compliance burden of this requirement by limiting information that should be disclosed to that which is "known or reasonably available," thereby eliminating any potential need for registrants

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[2769] We note that this "cost" is from the perspective of the ratings providers and could be offset by the efficiency gain that renders their intermediation less necessary; as such, it reflects more of a transfer than a net economic "cost."

[2770] *See* letter from the Heritage Foundation, which estimates compliance costs of the proposed rules on non-registrants would total $14 billion.

[2771] *See* letter from International Dairy Foods Association.

to undertake unreasonable searches or requests for information from such third parties. Given the more flexible and tailored approach in the final rules, such consultations will pertain only to parties whose relationship with the registrant is most likely to materially impact the registrant's strategy, business model and outlook, as well as parties from whom the registrant may be best positioned to request information, thus lowering these costs.

Some commenters asserted that a registrant's compliance costs could be passed on to other parties such as consumers (via higher prices), workers (through reduced wages or benefits), or shareholders (in the form of lower earnings).[2772] Other commenters stated that compliance costs could vary across industries.[2773] We acknowledge that third parties could bear some of the increased costs of compliance arising from the final rules and that this effect may be more pronounced in certain industries than in others. The final rules include significant changes from the proposal that lower the burdens on registrants. To the extent that these changes result in lower compliance costs, they also will help mitigate any adverse effects on other parties.

There is some existing academic literature on costs related to mandatory climate-related disclosures in other jurisdictions.[2774] Some studies report lower profitability and costly operational adjustments for firms affected by mandatory CSR disclosure and GHG emissions

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[2772] *See, e.g.*, letters from National Fuel Corporation; Petrol. OK; Footwear Distributors & Retails of America; Truth in Energy and Climate; ASA; and David R. Burton.

[2773] *See* letters from API; and Matthew Winden.

[2774] Commenters stated that there is limited evidence on the overall economic impact of mandatory climate-related disclosure regimes in other jurisdictions. *See* letter from Committee on Capital Markets Regulation (June 16, 2022) ("CCMR"); and Overdahl exhibit to letter from Chamber; *see also* Christensen *et al*. (2021).

reporting in China and the United Kingdom, respectively.[2775] However, other studies do not find an impact on financial operating performance from mandating climate-related disclosures.[2776] Another study showed aggregate stock price movement associated with mandatory climate-related disclosure; while the study found, on average, a negative market reaction, the negative stock returns were concentrated in firms with weak ESG performance and disclosure, while firms with above-median ESG performance and disclosure exhibited a positive abnormal return.[2777] We note that differences between the final rules and these other mandates (e.g., materiality qualifiers) suggest that similar costs associated with the final rules may be lower.

As discussed in sections IV.C.2.f and IV.C.3.c, the final rules may have implications for assurance providers or, more generally, for third parties with climate-related expertise. In the short run, the rules may increase demand (and accordingly, the cost) for climate-related expertise and/or assurance of emissions disclosures. Over time, we expect the supply of third parties with climate-related expertise will adjust to correspond with the increased demand, leading to reduced costs.

Finally, the modifications made in the final rules to reduce overall costs will help address, to an extent, some commenters' concerns that costs associated with the proposed rules could

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2775 *See* Y .Chen, M. Hung & Y. Wang, *The Effect of Mandatory CSR Disclosure on Firm Profitability and Social Externalities: Evidence from China*, 65 J. of Acct. & Econ. 169 (2018); *see also* letter from CCMR (citing, as evidence of negative effects to firm financial performance from mandatory climate-related disclosures, Jouvenot & P. Krueger, *Mandatory Corporate Carbon Disclosure: Evidence from a Natural Experiment* (Aug. 8, 2019), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3434490 (retrieved from SSRN Elsevier database)).

2776 *See* B. Downar, J. Ernstberger, S. Reichelstein, S. Schwenen & A. Zaklan, *The Impact of Carbon Disclosure Mandates on Emissions and Financial Operating Performance*, 26 Rev. of Acct. Stud. 1137 (2021).

2777 *See* letter from CCMR. *See also* Proposing Release, section IV.C.1., at n. 848, citing Jody Grewal, Edward J. Riedl & George Serafeim, *Market Reaction to Mandatory Nonfinancial Disclosure*, 65 Mgmt Sci. 3061 (2019). We note that the study's findings are based on the assumption that the only news disproportionately affecting the treated companies was the policy at issue, as opposed to some other event(s) impacting the treated companies.

factor into a company's decision to become or remain a public reporting company.[2778] In response to other commenters' concerns,[2779] the final rules also provide EGCs and SRCs with a longer phase in period for climate-related disclosures (including financial statement disclosures under Regulation S-X) and exempt EGCs and SRCs from GHG emissions disclosure requirements. And, while climate-related disclosures will be required in registration statements for firms conducting IPOs, we are not applying the subpart 1500 and Article 14 disclosure requirements to a private company that is a party to a business combination transaction, as defined by Securities Act Rule 165(f), involving a securities offering registered on Form S-4 or F-4.[2780]

2. Analysis of Specific Provisions

The costs incurred by any particular registrant may vary significantly depending upon which, if any, of the disclosures required under the final rules are applicable to that registrant's operations and circumstances. We discuss the costs of specific components of the rules below.

a. Disclosure of Climate-Related Risks

The final rules require registrants to identify any climate-related risks that have materially impacted or are reasonably likely to have a material impact on the registrant, including on its

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2778 *See, e.g.*, letter from Chamber and section II.L.2. For example, private companies might decide to defer a public offering, and existing public companies might decide to deregister from U.S. securities markets or not pursue mergers that would subject the merged company to reporting requirements. Other provisions that will reduce costs for conducting an IPO include (i) registrants will only have to provide Article 14 disclosure for historical fiscal years on a prospective basis, and (ii) the PSLRA statutory safe harbor for forward-looking statements (with respect to transition plans, scenario analysis, targets and goals) will apply to registration statements in IPOs.

2779 *See* sections II.H.2 and II.O.2 for a discussion of commenters' concerns on GHG emissions disclosures and phase in periods, respectively.

2780 *See supra* section II.L.3. While this approach avoids imposing additional costs on companies engaged in business combination transactions involving a private company, we note that investors will not have the benefit of the disclosures required by the final rules with respect to such private company.